Tactix latest-source proof and fuzz checkpoint

Codex for Andrew

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Outcome. The current source has a focused repair for a Sweep LRAT failure found by the older AM444 Opus campaign. A SAT-path public replay now checks every emitted LRAT lemma, whereas the old Release proof fails CakeLPR at line 509 and the old Debug solver aborts on an empty LRAT hint chain. Release and Debug focused tests each pass 4/4, including an original-CNF model and checker-backed regression. This is not a complete UNSAT-proof or N1–N4 acceptance result.

Reason for consolidating fuzz. AM427, AM444 and AM452 fuzzed separate immutable source snapshots. They were useful for finding C1, but no older result certifies the combined source. Following the request to fuzz only the latest version, 30 pending old-pin fuzz tasks were canceled, along with one pending old-source full Debug job. Already-running AM427 tasks may finish as diagnostics. The older AM444, AM452 and AM454 Opus owners were stopped; their preserved handoffs remain evidence. A new combined Opus campaign is required on the repaired source. The AM452 full Debug run timed out after 185 of 186 tests passed; its remaining demo needs a longer current-source allocation.

General repairs. AM444 C1 occurs when Sweep captures a binary direction proof, then an earlier root consequence fixes one endpoint. The emitted unit was sound, but its temporary LRAT lemmas were conditioned on a smaller cube than the one used to normalize their hints. The repair retains the already root-false endpoint in those temporary lemmas and their matching deletions. It changes neither Sweep's public action nor its final unit. The current source also includes AM454's proof-writer refinement: a rejected SR or external RUP addition does not close a pending H2 DSR-unit transaction. The canonical no-option native DSR-to-LSR-to-Trestle route and legal public X7 actions remain intact.

Evidence and limits. Job 1724272 reproduced the old Debug assertion; its old Release partial LRAT fails the pinned CakeLPR checker at line 509. Both repaired builds return SAT. An independent checker accepts the model against all 237 original clauses; CakeLPR checks every emitted LRAT line before reporting only the expected absence of a final empty clause on this SAT run. Job 1724291 passed the focused Release and Debug sets 4/4 each. AM452's exact 42-clause H2 public replay had already verified no-option dsr-trim in both directions and pinned Trestle on the original CNF, with identical Release and ASan proof bytes; AM454 preserved those bytes. The bounded AM444 API-UNSAT C1 case still needs a complete, independently checked proof on the current source. Combined Opus fuzz, full Debug, N1–N4 correctness and performance gates remain due.

Readiness and next actions. First verify the complete AM444 C1 UNSAT certificate, then run a single Opus-owned current-source campaign covering Sweep lifecycle, H1/C1, H2 and proof-writer boundaries with original-CNF models and independent certificates. Run the longer full Debug suite

and the remaining performance/reference gates. Online learning remains at prototype readiness: production needs adjustable inprocessing dose and order plus live owned switches for branching and clause deletion. Do not submit the 1,166-instance Tier D comparison until the goal is verified; its required limits remain 5,000 seconds, four CPUs and 32 GiB per instance on `cpu-dense-hwulab-q`.

Evidence. `benchmarks/reports/2026-09-25-am455-latest-source-fuzz-consolidation.md`; `benchmarks/reports/2026-09-25-am444-sweep-lrat-core-target-repair.md`; `benchmarks/reports/2026-09-25-am452-h2-dsr-unit-transaction.md`; `benchmarks/reports/2026-09-24-am376-online-learning-readiness.md`.